

Haskell.py Teacher's Edition

A Complete Course in Idiomatic Functional Python

nine chapters, 41 homework problems, 10 compare-and-contrast apps

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How to Take This Course

This book teaches `haskell.py` – `prelude.py`’s production sibling. Where `prelude.py` reimplements every tool from scratch for pedagogy, `haskell.py` WRAPS the standard library wherever it already has the tool (`scanl` is `itertools.accumulate`, `memo` is `functools.cache`) – one vocabulary, zero reimplementation. Failure is a value throughout: a true NOTHING sentinel keeps `None` legal as everyday data, parsers fail with `FAIL`, and `Either` carries `WHY`.

The whole library is one listing, up front, in Chapter 1 – read it once, top to bottom, the way you would read any short source file. The nine chapters after it are not a second pass over the same material: each opens with why its slice of the library exists and how its pieces fit together, then goes straight to **Homework** – drills cement each tool, applies combine them, and challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in; `python3 -m doctest` is the referee.

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Part I

The Nine Chapters

Chapter 1

haskell.py, Complete

Every name this book teaches, in one file, in the order it is defined – docstrings and comments stripped to the section headers, which are the only thing worth keeping as a wayfinding aid in a listing this short. Read it once before the first chapter; the chapters that follow assume you have.

```
1 from functools import reduce as _reduce, partial, cache as memo, cmp_to_key
2 from itertools import accumulate as _acc, islice as _islice, pairwise as _pairwise
3 from itertools import chain, count, cycle, repeat
4 from operator import add, sub, mul, not_
5 from math import gcd, lcm, hypot, isqrt, prod as product
6 from collections import Counter, defaultdict, deque
7 from heapq import heappush, heappop, merge as _hmerge
8 import re as _re
9 # 1. functions
10 identity = lambda x: x
11 const = lambda x: lambda _: x
12 flip = lambda f: lambda a,b: f(b,a)
13 curry = lambda f: lambda x: lambda y: f(x,y)
14 uncurry = lambda f: lambda p: f(*p)
15 def compose(*fs):
16     def g(x):
17         for f in reversed(fs):
18             x = f(x)
19         return x
20     return g
21 o = compose
22 def pipe(x,*fs):
23     for f in fs:
24         x = f(x)
25     return x
26 on = lambda f,g: lambda x,y: f(g(x),g(y))
27 # 2. pairs
28 fst = lambda p: p[0]
29 snd = lambda p: p[1]
30 swap = o(tuple,reversed)
31 # 3. sentinels
32 class _Sentinel:
33     __slots__ = ("_n",)
34     def __init__(self,n): self._n = n
35     def __repr__(self): return self._n
36     def __bool__(self): return False
37 NOTHING = _Sentinel("NOTHING")
38 FAIL = _Sentinel("FAIL")
39 _MISS = _Sentinel("_MISS")
40 # 4. arithmetic & logic
41 div = lambda a,b: a // b
42 mod = lambda a,b: a % b
43 halve = lambda n: n // 2
44 even = lambda n: n % 2 == 0
45 odd = lambda n: n % 2 == 1
46 otherwise = True
47 quot = lambda a,b: -(a // b) if (a < 0) != (b < 0) else a // b
48 rem = lambda a,b: a - b * quot(a,b)
49 quotRem = lambda a,b: (quot(a,b), rem(a,b))
50 signum = lambda x: (x > 0) - (x < 0)
51 succ = lambda x: chr(ord(x) + 1) if isinstance(x,str) else x + 1
```

```

52 pred = lambda x: chr(ord(x) - 1) if isinstance(x,str) else x - 1
53 # 5. folds, scans, unfolds
54 foldl = lambda f,xs,base=_MISS: _reduce(f,xs) if base is _MISS else _reduce(f,xs,base)
55 foldr = lambda f,xs,base: _reduce(flip(f),reversed(list(xs)),base)
56 foldl1 = foldl
57 foldr1 = lambda f,xs: _reduce(flip(f),reversed(list(xs)))
58 scanl = lambda f,xs,base=_MISS: list(_acc(xs,f) if base is _MISS else _acc(xs,f,initial=base))
59 scanl1 = scanl
60 scanr = lambda f,xs,base: scanl(flip(f),reversed(list(xs)),base)[::-1]
61 scanr1 = lambda f,xs: scanl(flip(f),reversed(list(xs)))[::-1]
62 def unfoldr(f,seed):
63     out = []
64     while (r := f(seed)) is not NOTHING:
65         out.append(r[0]); seed = r[1]
66     return out
67 def until(cond,f,x):
68     while not cond(x):
69         x = f(x)
70     return x
71 replicateM = lambda n,xs: foldl(lambda acc, _: [s + [x] for s in acc for x in xs],range(n),[[]])
72 subsequences = lambda xs: foldl(lambda acc,x: acc + [s + [x] for s in acc],list(xs),[[]])
73 # 6. streams
74 def iterate(f,x):
75     while True:
76         yield x
77         x = f(x)
78 take = lambda n,xs: list(_islice(iter(xs),n))
79 # 7. list basics
80 head = lambda xs: xs[0]
81 tail = lambda xs: xs[1:]
82 init = lambda xs: xs[:-1]
83 last = lambda xs: xs[-1]
84 drop = lambda n,xs: list(xs)[max(n,0):]
85 splitAt = lambda n,xs: (xs[:n], xs[n:]) if n >= 0 else (xs[:0], xs)
86 replicate = lambda n,x: [x] * n
87 elem = lambda x,xs: x in xs
88 notElem = lambda x,xs: x not in xs
89 null = lambda xs: len(xs) == 0
90 nub = o(list,dict.fromkeys)
91 enum = lambda xs,start=0: list(enumerate(xs,start))
92 pairwise = lambda xs: list(_pairwise(xs))
93 zip_ = lambda a,b: list(zip(a,b))
94 zip3 = lambda a,b,c: list(zip(a,b,c))
95 unzip = lambda ps: tuple(map(list,zip(*ps))) if ps else ([], [])
96 isPrefixOf = lambda p,xs: list(xs[:len(p)]) == list(p)
97 isSuffixOf = lambda p,xs: list(xs[len(xs) - len(p):]) == list(p)
98 lookup = lambda k,pairs: next((v for kk, v in pairs if kk == k),NOTHING)
99 stripPrefix = lambda p,xs: xs[len(p):] if isPrefixOf(p,xs) else NOTHING
100 # 8. higher-order lists
101 map_ = lambda f,xs: [f(x) for x in xs]
102 filter_ = lambda cond,xs: [x for x in xs if cond(x)]
103 concat = lambda xss: [x for xs in xss for x in xs]
104 concatMap = lambda f,xs: [y for x in xs for y in f(x)]
105 cross = lambda a,b: [(x, y) for x in a for y in b]
106 starmap = lambda f,pairs: [f(*p) for p in pairs]
107 zipWith = lambda f,a,b: [f(x,y) for x, y in zip(a,b)]
108 zipWith3 = lambda f,a,b,c: [f(x,y,z) for x, y, z in zip(a,b,c)]
109 find = lambda cond,xs: next((x for x in xs if cond(x)),NOTHING)
110 def partition(cond,xs):
111     yes, no = [], []
112     for x in xs:
113         (yes if cond(x) else no).append(x)
114     return (yes, no)
115 # 9. slicing & spans
116 def span(cond,xs):
117     xs = list(xs)
118     i = next((i for i, x in enumerate(xs) if not cond(x)),len(xs))
119     return (xs[:i], xs[i:])
120 break_ = lambda cond,xs: span(o(not_,cond),xs)
121 def takewhile(cond,xs):
122     for x in xs:
123         if not cond(x): return
124     yield x
125 def dropwhile(cond,xs):

```

```

126     it = iter(xs)
127     for x in it:
128         if not cond(x):
129             yield x
130             break
131     yield from it
132 def takeuntil(cond,xs):
133     for x in xs:
134         yield x
135         if cond(x): return
136 takeWhile = lambda cond,xs: list(takewhile(cond,xs))
137 dropWhile = lambda cond,xs: list(dropwhile(cond,xs))
138 inits = lambda xs: [xs[:i] for i in range(len(xs) + 1)]
139 tails = lambda xs: [xs[i:] for i in range(len(xs) + 1)]
140 def chunksOf(n,xs):
141     xs = list(xs)
142     return [xs[i:i + n] for i in range(0,len(xs),n)]
143 def windows(n,xs):
144     xs = list(xs)
145     return [xs[i:i + n] for i in range(len(xs) - n + 1)]
146 def groupBy(eq,xs):
147     out = []
148     for x in xs:
149         if out and eq(out[-1][0],x): out[-1].append(x)
150         else: out.append([x])
151     return out
152 group = partial(groupBy,lambda a,b: a == b)
153 def transpose(rows):
154     rows = [list(r) for r in rows]
155     out, i = [], 0
156     while any(i < len(r) for r in rows):
157         out.append([r[i] for r in rows if i < len(r)])
158         i += 1
159     return out
160 # 10. sorting & searching
161 sortOn = lambda f,xs: sorted(xs,key=f)
162 sortBy = lambda cmp,xs: sorted(xs,key=cmp_to_key(cmp))
163 maxOn = lambda f,xs: max(xs,key=f)
164 minOn = lambda f,xs: min(xs,key=f)
165 merge = lambda a,b: list(_hmerge(a,b))
166 # 11. strings
167 lines = str.splitlines
168 unlines = lambda ls: "".join(l + "\n" for l in ls)
169 words = str.split
170 unwords = " ".join
171 # 12. containers
172 def fromListWith(f,pairs):
173     d = {}
174     for k, v in pairs:
175         d[k] = f(v,d[k]) if k in d else v
176     return d
177 insertWith = lambda f,k,v,d: {**d, k: f(v,d[k]) if k in d else v}
178 unionWith = lambda f,a,b: {k: f(a[k],b[k]) if k in a and k in b else a[k] if k in a else b[k]
179                             for k in a.keys() | b.keys()}
180 def getpath(t,ks,default=NOTHING):
181     for k in ks:
182         if not isinstance(t,dict) or k not in t: return default
183         t = t[k]
184     return t
185 bag_diff = lambda a,b: {k: a[k] - b.get(k,0) for k in a if a[k] > b.get(k,0)}
186 bag_inter = lambda a,b: {k: v for k in a.keys() & b.keys() if (v := min(a[k],b[k]))}
187 bag_union = lambda a,b: {k: max(a.get(k,0),b.get(k,0)) for k in a.keys() | b.keys()}
188 bag_sub = lambda a,b: all(b.get(k,0) >= n for k,n in a.items())
189 # 13. Maybe
190 bind = lambda x,f: NOTHING if x is NOTHING else f(x)
191 isJust = lambda x: x is not NOTHING
192 isNothing = lambda x: x is NOTHING
193 fromMaybe = lambda default,x: default if x is NOTHING else x
194 listToMaybe = lambda xs: xs[0] if xs else NOTHING
195 maybeToList = lambda x: [] if x is NOTHING else [x]
196 catMaybes = lambda xs: [x for x in xs if x is not NOTHING]
197 mapMaybe = lambda f,xs: [y for y in map(f,xs) if y is not NOTHING]
198 def sequenceM(xs):
199     out = []

```

```

200     for x in xs:
201         if x is NOTHING: return NOTHING
202         out.append(x)
203     return out
204 traverseM = lambda f,xs: sequenceM(map_(f,xs))
205 maybe_get = lambda d,k: d.get(k,NOTHING)
206 # 14. Either
207 Ok = lambda v: ("ok", v)
208 Err = lambda e: ("err", e)
209 bindE = lambda x,f: x if x[0] == 'err' else f(x[1])
210 def sequenceE(xs):
211     out = []
212     for x in xs:
213         if x[0] == "err": return x
214         out.append(x[1])
215     return Ok(out)
216 traverseE = lambda f,xs: sequenceE(map_(f,xs))
217 def partitionEithers(xs):
218     oks, errs = [], []
219     for tag, v in xs:
220         (oks if tag == "ok" else errs).append(v)
221     return (oks, errs)
222 note = lambda msg,x: Err(msg) if x is NOTHING else Ok(x)
223 # 15. do-notation
224 def do(binder,is_fail):
225     def deco(gen_fn):
226         def run(*a,**kw):
227             g = gen_fn(*a,**kw)
228             def step(val):
229                 try:
230                     m = g.send(val)
231                     except StopIteration as e:
232                         return e.value
233                     return binder(m,step)
234             return step(None)
235         return run
236     return deco
237 doM = do(bind,lambda x: x is NOTHING)
238 doE = do(bindE,lambda x: x[0] == "err")
239 # 16. parser combinators
240 def doP(gen_fn):
241     def make(*a,**kw):
242         def parser(s):
243             g, val = gen_fn(*a,**kw), None
244             try:
245                 while True:
246                     r = g.send(val)(s)
247                     if r is FAIL: return FAIL
248                     val, s = r
249             except StopIteration as e:
250                 return (e.value, s)
251         return parser
252     return make
253 alt = lambda *ps: lambda s: next((r for r in (p(s) for p in ps) if r is not FAIL),FAIL)
254 def many(p):
255     def parser(s):
256         out = []
257         while (r := p(s)) is not FAIL and r[1] != s:
258             out.append(r[0]); s = r[1]
259         return (out, s)
260     return parser
261 def many1(p):
262     def parser(s):
263         r = p(s)
264         if r is FAIL: return FAIL
265         rest = many(p)(r[1])
266         return ([r[0]] + rest[0], rest[1])
267     return parser
268 def sepBy(p,sep):
269     def pair(s):
270         r = sep(s)
271         return FAIL if r is FAIL else p(r[1])
272     def parser(s):
273         r = p(s)

```

```

274         if r is FAIL: return ([], s)
275         rest = many(pair)(r[1])
276         return ([r[0]] + rest[0], rest[1])
277     return parser
278 def lit(c):
279     def parser(s):
280         s = s.lstrip()
281         return (c, s[1:]) if s[:1] == c else FAIL
282     return parser
283 def rx(pat,conv=identity):
284     r = _re.compile(pat)
285     def parser(s):
286         s = s.lstrip()
287         m = r.match(s)
288         return (conv(m.group()), s[m.end():]) if m else FAIL
289     return parser
290 def chainl1(p,ops):
291     opp = alt(*[lit(c) for c in ops])
292     def parser(s):
293         vr = p(s)
294         while vr is not FAIL:
295             op = opp(vr[1])
296             if op is FAIL: return vr
297             w = p(op[1])
298             if w is FAIL: return vr
299             vr = (ops[op[0]](vr[0],w[0]), w[1])
300         return FAIL
301     return parser
302 def runParser(p,s):
303     r = p(s)
304     if r is FAIL: return Err(f"no parse: {s!r}")
305     if r[1].strip(): return Err(f"unconsumed input: {r[1].strip()!r}")
306     return Ok(r[0])
307 # 17. monoids
308 Monoid = lambda empty,op: (empty, op)
309 mconcat = lambda m,xs: foldl(m[1],xs,m[0])
310 foldMap = lambda f,m,xs: mconcat(m,[f(x) for x in xs])
311 both = lambda m1,m2: Monoid((m1[0], m2[0]),
312     lambda a,b: (m1[1](a[0],b[0]), m2[1](a[1],b[1])))
313 Sum = Monoid(0,add)
314 Product = Monoid(1,mul)
315 All = Monoid(True,lambda a,b: a and b)
316 Any = Monoid(False,lambda a,b: a or b)
317 MaxM = Monoid(float("-inf"),max)
318 MinM = Monoid(float("inf"),min)
319 ListM = Monoid([],add)
320 First = Monoid(NOTHING,lambda a,b: b if a is NOTHING else a)
321 Last = Monoid(NOTHING,lambda a,b: a if b is NOTHING else b)

```


Chapter 2

Point-Free Foundations

A function that names its argument commits to one shape of input; a function built from other functions (`compose`, `on`, `partial`) stays a transformation, reusable wherever the types line up. This chapter’s five one-liners (`identity`, `const`, `flip`, `curry`, `uncurry`) are the adapters that make composition possible at all – they change nothing about the values, only the SHAPE of the call. Two sentinels close the chapter: `NOTHING` means “no answer”, and `FAIL` means “the parser gave up” – both real objects, never `None`, so a function can return an ordinary `None` as genuine data without it being mistaken for absence.

2.1 Homework

Work in order: drills cement each tool, applies combine them, challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in, and let `python3 -m doctest` referee.

1.1 Mirror a pair

drill

Write `mirror(p)`: the pair with its elements exchanged, point-free. Applying it twice must give the original back.

```
mirror(p) -> pair
|>>> mirror((1, 2))
| (2, 1)
|>>> mirror(mirror((3, 4)))
| (3, 4)
```

Solution.

```
| mirror = o(tuple, reversed)
```

Note. Reads as a sentence: tuple of reversed. No argument ever named.

1.2 Record projections

drill

A leaderboard record is a (name, score) pair. Write `name_of` and `score_of` point-free, without lambdas.

```
name_of(r) -> value; score_of(r) -> value
|>>> name_of(('ada', 95))
| 'ada'
|>>> score_of(('ada', 95))
| 95
```

Solution.

```
| name_of, score_of = itemgetter(0), itemgetter(1)
```

Note. `operator.itemgetter` is Python’s `fst/snd` – and it slots straight into `map` and `sorted(key=...)`.

1.3 Default when missing

drill

Write `port_of(cfg)`: the 'port' entry of a config dict, defaulting to 8080 only when the key is absent. A key holding `None` is real data and must survive.

`port_of(cfg) -> value`

```
>>> port_of({'port': 9000})
9000
>>> port_of({})
8080
>>> port_of({'port': None}) is None
True
```

Solution.

```
| port_of = lambda cfg: fromMaybe(8080, maybe_get(cfg, 'port'))
```

Note. The prelude course's `or_default` cannot pass the third test: with `None-as-Nothing`, `None` IS the miss. The `NOTHING` sentinel is the fix.

1.4 Clamp to a range

applied

Write `clamp(lo, hi)`: a FUNCTION that forces a number into `[lo, hi]`. Build it from `max` and `min` with partial application – no `lambda` in the returned function.

`clamp(lo, hi) -> (x -> value)`

```
>>> clamp(0, 10)(15)
10
>>> clamp(0, 10)(-5)
0
>>> clamp(0, 10)(7)
7
```

Solution.

```
| clamp = lambda lo, hi: o(partial(max, lo), partial(min, hi))
```

Note. Right to left: cap above with `min(hi, .)`, then floor with `max(lo, .)`. Sections via `partial`.

1.5 Twice

drill

Write `twice(f)`: the function that applies `f` two times.

`twice(f) -> (x -> value)`

```
>>> twice(lambda n: n * 3)(2)
18
>>> twice(str.strip)(' hi ')
'hi'
```

Solution.

```
| twice = lambda f: o(f, f)
```

Note. `twice = o` on the diagonal. `twice(twice(f))` is `f` four times – combinators compose with themselves.

Chapter 3

Folds, Scans and Unfolds

A fold collapses a list to one value; a scan is a fold that keeps its history; an unfold grows a list from a seed instead of consuming one. `scanl` here is *exactly* `itertools.accumulate` – naming it the Haskell way does not change what runs. `unfoldr` is the fold’s mirror image: given a step function that returns `(value, next_seed)` or `NOTHING` to stop, it produces the whole list, which is why balances, digit sequences and Collatz chains all turn out to be one `unfoldr` call apiece. `iterate` is `unfoldr`’s lazy, infinite cousin: it never stops on its own, so `take` is what makes it usable.

3.1 Homework

Work in order: drills cement each tool, applies combine them, challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in, and let `python3 -m doctest referee`.

2.1 Parity census

drill

Count evens and odds as a pair (evens, odds), in ONE pass, point-free at the top level.

`parity_counts(xs) -> (evens, odds)`

```
>>> parity_counts([1, 2, 3, 4])
(2, 2)
>>> parity_counts([])
(0, 0)
>>> parity_counts([-2, -3])
(1, 1)
```

Solution.

```
| parity_counts = partial(foldMap, lambda x: (1 - x % 2, x % 2), both(Sum, Sum))
```

Note. `both(Sum, Sum)` is two counters riding one fold; `partial` freezes the fold into a name. The `[]` case is free: monoid identity.

2.2 Running balance

drill

An account starts at 0; deposits are positive, withdrawals negative. Return every intermediate balance including the start.

`balances(txs) -> [balance]`

```
>>> balances([100, -30, -30, 5])
[0, 100, 70, 40, 45]
>>> balances([])
[0]
```

Solution.

```
| balances = lambda txs: scanl(add, txs, 0)
```

Note. `Scan` = fold that keeps its diary. `operator.add` instead of a lambda.

2.3 Countdown

drill

Write `countdown(n)`: the list `n, n-1, ..., 1` grown by `unfoldr`, point-free at the top level.

`countdown(n) -> [int]`

```
>>> countdown(4)
[4, 3, 2, 1]
>>> countdown(0)
[]
```

Solution.

```
countdown = partial(unfoldr, lambda n: NOTHING if n == 0 else (n, n - 1))
```

Note. `unfoldr` is the anamorphism: seed in, list out, `NOTHING` stops it.

2.4 Digits of a number

applied

Write `digits(n)`: the base-10 digits of a non-negative integer, most significant first. Zero has one digit.

`digits(n) -> [int]`

```
>>> digits(407)
[4, 0, 7]
>>> digits(0)
[0]
```

Solution.

```
digits = lambda n: unfoldr(lambda k: NOTHING if k == 0 else (k % 10, k // 10), n)[::-1] or [0]
```

Note. `Unfold` peels least-significant first; `reverse` restores reading order. `'or [0]'` patches the empty `unfold` at zero.

2.5 Collatz flight length

challenge

Write `collatz_len(n)`: how many numbers the Collatz sequence visits from `n` down to 1, inclusive. Even halves; odd triples-plus-one.

`collatz_len(n) -> int`

```
>>> collatz_len(6)
9
>>> collatz_len(1)
1
```

Solution.

```
collatz_len = lambda n: 1 + len(unfoldr(lambda k: NOTHING if k == 1 else (k, k // 2 if k % 2 == 0 else 3 * k + 1), n))
```

Note. The `unfold` emits everything before the fixed point; `+1` counts the 1 itself. No mutation, no `while`.

Chapter 4

The List Gaps

The largest chapter because it is the least glamorous: the ordinary list and dict work that every program needs, named consistently instead of reached for ad hoc. `windows` and `groupBy` both slide a view across a sequence, but for different reasons – one for a fixed-width lookback (moving averages), one for runs of equal elements (RLE, compression). `transpose` is ragged-safe on purpose: short rows simply drop out of later columns instead of raising, so uneven data never needs a special case. `fromListWith` and `unionWith` are the two dict-building folds worth memorising – almost every “group these” or “merge these two counters” problem is one of them with the right combining function.

4.1 Homework

Work in order: drills cement each tool, applies combine them, challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in, and let `python3 -m doctest referee`.

3.1 Run-length encoding

drill

Compress a sequence into (element, count) pairs per maximal run.

```
rle(xs) -> [(x, n)]

>>> rle('aaabbc')
[('a', 3), ('b', 2), ('c', 1)]
>>> rle([])
[]
```

Solution.

```
rle = lambda xs: [(g[0], len(g)) for g in groupBy(eq, xs)]
```

Note. `groupBy(eq, .)` does the segmentation; the comprehension just labels each run. `operator.eq`, no `lambda`.

3.2 Moving average

applied

Write `mavg(n, xs)`: the `n`-point moving average of a signal.

```
mavg(n, xs) -> [float]

>>> mavg(3, [1, 2, 3, 4, 5])
[2.0, 3.0, 4.0]
>>> mavg(3, [1, 2])
[]
```

Solution.

```
mavg = lambda n, xs: [sum(w) / n for w in windows(n, xs)]
```

Note. `windows` names the sliding view; the arithmetic stays arithmetic. Too-short input yields no windows – correct for free.

3.3 Pagination

drill

Write `paginate(n, xs)`: pages of at most `n` items, last page short.

```
paginate(n, xs) -> [[x]]
```

```
>>> paginate(2, [1, 2, 3, 4, 5])
[[1, 2], [3, 4], [5]]
```

Solution.

```
| paginate = chunksOf
```

Note. Recognize-it drill: pagination IS `chunksOf`.

3.4 Column sums, ragged

applied

Sensor rows arrive with trailing dropouts. Sum each column over the rows that reached it – point-free.

```
colsums(rows) -> [sum]
```

```
>>> colsums([[1, 2, 3], [4, 5]])
[5, 7, 3]
>>> colsums([])
[]
```

Solution.

```
| colsums = o(list, partial(map, sum), transpose)
```

Note. Ragged-safe transpose turns dropouts into short columns instead of exceptions; then it is just `map sum`.

3.5 Indentation split

drill

Split a line into (leading whitespace, rest) as strings, one pass.

```
indent_split(s) -> (str, str)
```

```
>>> indent_split('  x = 1')
('  ', 'x = 1')
>>> indent_split('top')
('', 'top')
```

Solution.

```
| indent_split = lambda s: tuple(''.join(p) for p in span(str.isspace, s))
```

Note. `span` is the two-sided `takewhile`; the `join` restores strings from character lists.

Chapter 5

Maybe, Done Right

`None` is Python’s “no value”, which is exactly the problem: a function cannot then return `None` to mean a REAL answer without creating an ambiguity. `NOTHING` fixes this by being a distinct object that is never a legitimate value – `maybe_get` returns `None` for a key whose stored value truly is `None`, and `NOTHING` only when the key is missing outright. `bind` chains failable steps by short-circuiting the moment one returns `NOTHING`; `sequenceM` and `traverseM` extend that to a whole list, all-or-nothing. Once a pipeline’s failure mode is a value instead of an exception, the pipeline composes like any other function.

5.1 Homework

Work in order: drills cement each tool, applies combine them, challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in, and let `python3 -m doctest referee`.

4.1 Safe head

drill

Write `headM(xs)`: the first element, or `NOTHING` for an empty sequence. A leading `None` is data.

`headM(xs) -> Maybe value`

```
>>> headM([7, 8])
7
>>> headM([]) is NOTHING
True
>>> headM([None]) is None
True
```

Solution.

```
headM = lambda xs: xs[0] if xs else NOTHING
```

Note. The third test is the whole point of the sentinel: `None` rides through as a value.

4.2 Deep dictionary path

applied

Write `pathM(d, ks)`: follow a key path through nested dicts; `NOTHING` if any hop is missing. A stored `None` must survive the trip.

`pathM(d, ks) -> Maybe value`

```
>>> d = {'db': {'host': 'local', 'pool': None}}
>>> pathM(d, ['db', 'host'])
'local'
>>> pathM(d, ['db', 'pool']) is None
True
>>> pathM(d, ['db', 'user']) is NOTHING
True
```

Solution.

```
pathM = lambda d, ks: foldl(lambda m, k: bind(m, lambda t: maybe_get(t, k)), ks, d)
```

Note. A path lookup is a FOLD of binds over the keys – the `Maybe` monad’s signature move.

4.3 Salvage the parseable

drill

Given strings, keep the ones that are integers, as ints, dropping the rest.

```
salvage_ints(ss) -> [int]
```

```
>>> salvage_ints(['3', 'x', '-7', ''])
[3, -7]
```

Solution.

```
salvage_ints = lambda ss: catMaybes([int(s) if s.lstrip('-').isdigit() and s else NOTHING for s in ss])
```

Note. Map into Maybe, then catMaybes sweeps the failures out – mapMaybe in two named steps.

4.4 All keys or nothing

applied

Write need(d, ks): the values for ks in order, or NOTHING if ANY key is absent.

```
need(d, ks) -> Maybe [value]
```

```
>>> need({'a': 1, 'b': 2}, ['a', 'b'])
[1, 2]
>>> need({'a': 1}, ['a', 'b']) is NOTHING
True
```

Solution.

```
need = lambda d, ks: sequenceM([maybe_get(d, k) for k in ks])
```

Note. sequenceM is the all-or-nothing gate: one miss poisons the batch.

4.5 First responder

challenge

Several config sources; write first_set(dict, k): the value from the FIRST dict that defines k at all – even as None – else NOTHING.

```
first_set(dict, k) -> Maybe value
```

```
>>> first_set([{}], {'x': None}, {'x': 3}, 'x') is None
True
>>> first_set([{}], {}, 'x') is NOTHING
True
```

Solution.

```
first_set = lambda dict, k: mconcat(First, [maybe_get(d, k) for d in dict])
```

Note. First’s identity is NOTHING, so ‘defined as None’ beats ‘undefined’ – exactly the override semantics config layering needs.

Chapter 6

Either: Failure With a Why

Maybe answers “did it work?”; Either answers that AND “why not?”. `Ok/Err` are tagged pairs – (“ok”, `v`) or (“err”, `why`) – so `bindE` can short-circuit exactly like `bind` while still carrying the failure’s reason to wherever the pipeline is finally inspected. `sequenceE` and `traverseE` are Either’s all-or-nothing gates, and `note` is the bridge from the last chapter: it turns a `NOTHING` into an `Err` with a message attached, for the moment a Maybe pipeline needs to explain itself to a caller.

6.1 Homework

Work in order: drills cement each tool, applies combine them, challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in, and let `python3 -m doctest referee`.

5.1 Lift the miss into a reason

drill

Write `get_e(d, k)`: `Ok(value)` or `Err` naming the missing key.

`get_e(d, k) -> Either`

```
>>> get_e({'a': 1}, 'a')
('ok', 1)
>>> get_e({}, 'user')
('err', "missing key: 'user'")
```

Solution.

```
get_e = lambda d, k: note(f"missing key: {k!r}", maybe_get(d, k))
```

Note. `note` is the Maybe-to-Either bridge: same short-circuit, now with a why.

5.2 Parse then bound

applied

Write `to_port(s)`: parse a decimal string and demand 1..65535, each failure carrying its own message.

`to_port(s) -> Either`

```
>>> to_port('8080')
('ok', 8080)
>>> to_port('http')
('err', "not a number: 'http'")
>>> to_port('99999')
('err', 'out of range: 99999')
```

Solution.

```
to_port = lambda s: bindE(
    note(f"not a number: {s!r}", int(s) if s.isdigit() else NOTHING),
    lambda n: Ok(n) if 1 <= n <= 65535 else Err(f"out of range: {n}"))
```

Note. Two failure modes, two messages, one `bindE`. The caller pattern-matches once.

5.3 First error speaks

drill

Validate a batch: all Ok values, or the FIRST Err verbatim.

`all_ports(ss) -> Either`

```
>>> all_ports(['80', '443'])
('ok', [80, 443])
>>> all_ports(['80', 'x', 'y'])
('err', "not a number: 'x'")
```

Solution.

```
all_ports = lambda ss: sequenceE([to_port(s) for s in ss])
```

Note. `sequenceE` = `sequenceM` that can explain itself.

5.4 Division with a why

drill

Write `div_e(a, b)`: the quotient, or an error naming the offense.

`div_e(a, b) -> Either`

```
>>> div_e(10, 4)
('ok', 2.5)
>>> div_e(1, 0)
('err', 'division by zero')
```

Solution.

```
div_e = lambda a, b: Err('division by zero') if b == 0 else Ok(a / b)
```

Note. Exceptions escape; Either values travel. Choose Either at module boundaries.

Chapter 7

do-Notation

Chaining `bind` calls by hand nests one call inside the next, one level per step – the “bind pyramid”. `do` rebuilds that same chain out of an ordinary generator function: each `yield` is one `bind`, and the value sent back in is what the step would have returned. `doM` wires this to the Maybe monad, `doE` to Either – same decorator, same generator shape, different short-circuit rule. The win is not cleverness for its own sake: a five-step Maybe pipeline written with `doM` reads top to bottom, in the order the steps actually happen, with the failure path made invisible instead of handled at every line.

7.1 Homework

Work in order: drills cement each tool, applies combine them, challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in, and let `python3 -m doctest` referee.

6.1 Config reader, flat

drill

Using `doM`, read `host` and `port` from a dict; NOTHING if either is absent.

`hostport(d) -> Maybe (host, port)`

```
>>> hostport({'host': 'h', 'port': 80})
('h', 80)
>>> hostport({'host': 'h'}) is NOTHING
True
```

Solution.

```
@doM
def hostport(d):
    h = yield maybe_get(d, 'host')
    p = yield maybe_get(d, 'port')
    return (h, p)
```

Note. Compare the bind pyramid this replaces: each `yield` is one `»=`, and the failure path is invisible.

6.2 Join across two tables

applied

`users` maps `id` to `name`; `emails` maps `name` to `address`. Using `doM`, resolve an `id` to its address.

`address(users, emails, uid) -> Maybe str`

```
>>> u, e = {1: 'ada'}, {'ada': 'ada@x.io'}
>>> address(u, e, 1)
'ada@x.io'
>>> address(u, e, 2) is NOTHING
True
```

Solution.

```
@doM
def address(users, emails, uid):
```

```

    name = yield maybe_get(users, uid)
    return (yield maybe_get(emails, name))

```

Note. A two-hop join is two yields. The second bind consumes the first's value by name, not by nesting.

6.3 Validated record

applied

Using doE, build a user record from a form dict: name must be non-empty, age must parse and land in 0..130. Each rejection explains itself.

mk_user(form) -> Either dict

```

>>> mk_user({'name': 'ada', 'age': '36'})
('ok', {'name': 'ada', 'age': 36})
>>> mk_user({'name': '', 'age': '36'})
('err', 'empty name')
>>> mk_user({'name': 'ada', 'age': 'old'})
('err', "bad age: 'old'")

```

Solution.

```

@doE
def mk_user(form):
    name = yield (Ok(form.get('name', '')) if form.get('name') else Err('empty name'))
    raw = form.get('age', '')
    age = yield note(f"bad age: {raw!r}", int(raw) if raw.isdigit() else NOTHING)
    yield (Ok(age) if age <= 130 else Err(f'implausible age: {age}'))
    return Ok({'name': name, 'age': age})

```

Note. A bare yield with no binding is a guard: run the check, keep no value. The happy path reads top to bottom.

6.4 Chained arithmetic

drill

Using doE and div_e from 5.4, compute $a/b + c/d$ with the first bad divisor reported.

sum_of_ratios(a, b, c, d) -> Either

```

>>> sum_of_ratios(1, 2, 1, 4)
('ok', 0.75)
>>> sum_of_ratios(1, 0, 1, 4)
('err', 'division by zero')

```

Solution.

```

@doE
def sum_of_ratios(a, b, c, d):
    x = yield div_e(a, b)
    y = yield div_e(c, d)
    return Ok(x + y)

```

Note. Error plumbing gone; only the arithmetic remains.

Chapter 8

Parser Combinators

A parser is a function from a string to `(value, rest)` or `FAIL` – once that shape is fixed, parsers compose like any other function. `doP` threads the remaining input through a generator the same way `doM` threads a `Maybe`; `alt` tries alternatives in order; `many` and `sepBy` handle repetition and separator-delimited lists without recursion. `chainl1` deserves special attention: it folds a sequence of `p (op p) * LEFT`, which is exactly how left-associative arithmetic is supposed to parse, and it is the one combinator in this chapter doing real algorithmic work rather than just gluing others together.

8.1 Homework

Work in order: drills cement each tool, applies combine them, challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in, and let `python3 -m doctest referee`.

7.1 Comma-separated integers

drill

Parse a comma-separated integer list, whitespace-blind, whole input consumed.

`int_list(s) -> Either [int]`

```
>>> int_list('1, -2 , 3')
('ok', [1, -2, 3])
>>> int_list('1, x')
('err', "unconsumed input: ' , x'")
```

Solution.

```
int_list = lambda s: runParser(sepBy(rx(r' -?\d+', int), lit(',')), s)
```

Note. `sepBy` handles the fencepost (separators BETWEEN items); `runParser` demands the input be spent and lands in `Either`.

7.2 key=value line

drill

Parse one 'key = value' assignment into a pair; keys are words, values run to end of line.

`kv(s) -> Either (k, v)`

```
>>> kv('host = 10.0.0.1')
('ok', ('host', '10.0.0.1'))
>>> kv(' = oops')[0]
'err'
```

Solution.

```
@doP
def _kv():
    k = yield rx(r'\w+')
    yield lit('=')
    v = yield rx(r'[\n]+' , str.strip)
    return (k, v)
kvline = _kv()
kv = lambda s: runParser(kvline, s)
```

Note. Three yields, no rest-threading anywhere. kvline is reusable – chapter 9’s INI reader is built on it.

7.3 CSV row

drill

Split one unquoted CSV row into stripped fields.

```
csv_row(s) -> Either [str]
|>>> csv_row('ada, 95, pass')
|('ok', ['ada', '95', 'pass'])
```

Solution.

```
| csv_row = lambda s: runParser(sepBy(rx(r'[^,\n]+'), str.strip), lit(','), s)
```

Note. The same sepBy skeleton as 7.1 with a different token – combinator reuse is the whole pitch.

7.4 ISO date

applied

Parse YYYY-MM-DD into an (y, m, d) integer triple, rejecting month 13.

```
iso_date(s) -> Either (y, m, d)
|>>> iso_date('2026-09-17')
|('ok', (2026, 9, 17))
|>>> iso_date('2026-13-01')
|('err', 'bad month: 13')
```

Solution.

```
num = lambda n: rx(r'\d{%d}' % n, int)
@doP
def _date():
    y = yield num(4)
    yield lit('-')
    m = yield num(2)
    yield lit('-')
    d = yield num(2)
    return (y, m, d)
iso_date = lambda s: bindE(runParser(_date(), s),
    lambda t: Ok(t) if 1 <= t[1] <= 12 else Err(f'bad month: {t[1]}'))
```

Note. Parse the shape with doP, judge the meaning with bindE – syntax and semantics in separate layers.

7.5 Sums and differences

challenge

Parse and evaluate a left-associative +/- chain over integers.

```
sums(s) -> Either int
|>>> sums('10 - 3 - 2 + 1')
|('ok', 6)
|>>> sums('10 -')[0]
|'err'
```

Solution.

```
| sums = lambda s: runParser(chainl1(rx(r'[-?\d+]', int), {'+': add, '-': sub}), s)
```

Note. chainl1 IS left-associativity-as-a-fold; operator.add/sub keep it point-free. Chapter 9 grows this into the full calculator.

Chapter 9

Monoids at Work

A monoid is nothing but an identity element paired with an associative combiner, reified as the pair `(empty, op)` – naming it as `DATA` means one engine, `mconcat`, folds every instance, and `foldMap` fuses “measure each element” with “combine the measurements” into one pass. `both` is the chapter’s real payoff: it pairs two monoids into one, so a max and a min, or a count and a total, fall out of a `SINGLE` traversal instead of two separate loops. That matters most exactly when it looks like it should not – a one-shot iterator or a huge file that can only be read once.

9.1 Homework

Work in order: drills cement each tool, applies combine them, challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in, and let `python3 -m doctest referee`.

8.1 Min and max, one pass

drill

Both extremes of a numeric stream in a single traversal, point-free.

```
extremes(xs) -> (min, max)
```

```
>>> extremes([3, 1, 4, 1, 5])
(1, 5)
```

Solution.

```
extremes = partial(foldMap, lambda x: (x, x), both(MinM, MaxM))
```

Note. `both` fuses two folds into one pass – what matters when `xs` is a one-shot iterator or a large file.

8.2 Stream statistics

applied

Count, total and maximum of a stream in ONE pass: `((count, total), maximum)`.

```
stats(xs) -> ((n, total), mx)
```

```
>>> stats([3, 1, 4])
((3, 8), 4)
```

Solution.

```
stats = partial(foldMap, lambda x: ((1, x), x), both(both(Sum, Sum), MaxM))
```

Note. `both` nests: a monoid of monoids. The tuple shape of the answer mirrors the shape of the fold.

8.3 Config layering

drill

defaults, file, cli – highest layer that `DEFINES` a key wins, and defining it as `None` counts. Point-free over the layer list.

```
effective(layers, k) -> Maybe value
```

```
>>> layers = [{'v': 1}, {}, {'v': None}]
>>> effective(layers, 'v') is None
```

```
True
>>> effective(layers, 'x') is NOTHING
True
```

Solution.

```
effective = lambda layers, k: mconcat>Last, [maybe_get(d, k) for d in layers])
```

Note. Last with a NOTHING identity is override semantics verbatim. Same skeleton as 4.5 with the opposite monoid.

8.4 Flatten

drill

Concatenate a list of lists by monoid fold.

```
flatten(xss) -> [x]
```

```
>>> flatten([[1], [], [2, 3]])
[1, 2, 3]
```

Solution.

```
flatten = partial(mconcat, ListM)
```

Note. Recognize-it: concatenation is the list monoid's mconcat. (For huge inputs reach for itertools.chain – this one is quadratic.)

Chapter 10

Capstone Labs

No new vocabulary – this chapter is the payoff for the previous eight. Each problem reaches back for tools from more than one chapter at once: parsing with `doP`, then judging the parsed value with `bindE`; folding a monoid across a batch validated with `doE`. If a chapter's tools felt like isolated tricks on first reading, this is where they stop being isolated.

10.1 Homework

Work in order: drills cement each tool, applies combine them, challenges are interview-grade. Write each solution in a scratch file with the given doctests pasted in, and let `python3 -m doctest referee`.

9.1 The algebraic calculator

lab

Full expression evaluator: `+` `-` `*` `/`, precedence, LEFT associativity, unary minus, parentheses, decimals, whitespace; Either out.

`calc(s) -> Either float`

```
>>> calc('2 * (3 + 4) - -5')
('ok', 19.0)
>>> calc('8 - 3 - 2')
('ok', 3.0)
>>> calc('2 * (3 +)')[0]
'err'
```

Solution.

```
number = rx(r'[-?\d+(\.\d+)?:', float)
@doP
def _paren():
    yield lit('(')
    v = yield expr
    yield lit(')')
    return v
unary = lambda s: alt(number, _paren())(s)
term = chain1(unary, {'*': mul, '/': lambda a, b: a / b})
expr = chain1(term, {'+': add, '-': sub})
calc = partial(runParser, expr)
```

Note. Eight grammar lines. Every layer earns its keep: `doP` threads state, `chain1` folds associativity, `runParser` explains failure. 5000-term chains run in constant stack.

9.2 INI reader

lab

Parse `[section]` headers with `key = value` bodies into a dict of dicts.

`ini(s) -> Either section: k: v`

```
>>> t = '[db]\nhost = local\nport = 5432\n[app]\ndebug = true'
>>> ini(t)
('ok', {'db': {'host': 'local', 'port': '5432'}, 'app': {'debug': 'true'}})
```

Solution.

```

@doP
def _kvi():
    k = yield rx(r'\w+')
    yield lit('=')
    v = yield rx(r'^\n\[ ]+', str.strip)
    return (k, v)
@doP
def _section():
    yield lit('[')
    name = yield rx(r'\w+')
    yield lit(']')
    kvs = yield many(_kvi())
    return (name, dict(kvs))
ini = lambda s: bindE(runParser(many(_section()), s), o(Ok, dict))

```

Note. The 7.2 skeleton, nested once. many loops, so thousand-key files cost no stack; bindE lifts the final dict without unwrapping by hand.

9.3 Log triage*lab*

Lines are 'LEVEL message'. In ONE pass report ((errors, total), first ERROR message or NOTHING).

trriage(lines) -> ((errors, total), first_err)

```

>>> triage(['INFO up', 'ERROR db down', 'ERROR retry', 'INFO ok'])
((2, 4), 'db down')
>>> triage(['INFO up'])[1] is NOTHING
True

```

Solution.

```

tag = lambda ln: (lambda lvl, _, msg: ((lvl == 'ERROR', 1),
                                     msg if lvl == 'ERROR' else NOTHING))(*ln.partition(' '))
trriage = partial(foldMap, tag, both(both(Sum, Sum), First))

```

Note. Counting and first-witness are DIFFERENT monoids riding the same pass. Adding a new statistic is one more both, not another loop.

9.4 Batch admission*lab*

Admit a batch of form dicts with 6.3's mk_user: all records, or the first rejection verbatim.

admit(forms) -> Either [dict]

```

>>> admit([{'name': 'ada', 'age': '36'}, {'name': 'bob', 'age': '41'}])
('ok', [{'name': 'ada', 'age': 36}, {'name': 'bob', 'age': 41}])
>>> admit([{'name': 'ada', 'age': '36'}, {'name': '', 'age': '2'}])
('err', 'empty name')

```

Solution.

```

admit = o(sequenceE, partial(map, mk_user), list)

```

Note. Validation composes: per-record doE, batch-level sequenceE, glued point-free. The whole intake pipeline is one line.

Part II

Compare and Contrast: Imperative vs Functional

About this part

Ten small apps, each written twice over the same inputs: once imperative, once with `haskell.py`, with a `__main__` block that ASSERTS the two agree. Every listing below is verified to actually run and agree before this book is built – what you read is what was executed.

	imperative	functional
state	mutable slots you maintain	threaded by the combinator
errors	exceptions / early return	values (NOTHING, Err) that compose
no value yet	hand-rolled sentinel per site	NOTHING, once, in the library
passes over data	one loop per concern	monoids fuse concerns into one pass
extension	edit inside the loop	add a stage / another both

Chapter 11

calculator

evaluate '2 * (3 + 4) - 5' both ways

Imperative

```
def calc_imp(s):
    toks, i = [], 0
    while i < len(s):
        c = s[i]
        if c.isspace(): i += 1; continue
        if c in '+-*/()': toks.append(c); i += 1; continue
        if c.isdigit():
            j = i
            while j < len(s) and (s[j].isdigit() or s[j] == '.'): j += 1
            toks.append(float(s[i:j])); i = j; continue
        return ('err', f'bad char {c!r}')

class P:
    pos = 0
def peek(): return toks[P.pos] if P.pos < len(toks) else None
def unary():
    t = peek()
    if t == '-': P.pos += 1; return -unary()
    if t == '(':
        P.pos += 1; v = expr()
        if peek() != ')': raise ValueError('expected ')
        P.pos += 1; return v
    if isinstance(t, float): P.pos += 1; return t
    raise ValueError(f'unexpected {t!r}')
def term():
    v = unary()
    while peek() in ('*', '/'):
        op = peek(); P.pos += 1; w = unary()
        v = v * w if op == '*' else v / w
    return v
def expr():
    v = term()
    while peek() in ('+', '-'):
        op = peek(); P.pos += 1; w = term()
        v = v + w if op == '+' else v - w
    return v

try:
    v = expr()
    return ('ok', v) if P.pos == len(toks) else ('err', 'unconsumed input')
except ValueError as e:
    return ('err', str(e))
```

Functional

```
number = rx(r'-?\d+(\.\d+)?', float)
@doP
def _paren():
    yield lit('(')
    v = yield expr_fn
    yield lit(')')
    return v
unary_fn = lambda s: alt(number, _paren())(s)
term_fn = chainl1(unary_fn, {'*': mul, '/': lambda a, b: a / b})
expr_fn = chainl1(term_fn, {'+': add, '-': sub})
calc_fn = partial(runParser, expr_fn)
```

Chapter 12

gradebook

validate 'name,score' rows, then one-pass class stats

Imperative

```
def admit_imp(rows):
    out = []                                # accumulate + early return
    for row in rows:
        name, _, raw = row.partition(',')
        if not name:
            return ('err', f'empty name in {row!r}')
        if not raw.strip().isdigit():
            return ('err', f'bad score in {row!r}')
        score = int(raw)
        if score > 100:
            return ('err', f'score > 100 in {row!r}')
        out.append((name, score))
    n = tot = 0                             # second loop for stats
    best = float('-inf')
    for _, s in out:
        n += 1; tot += s
        if s > best: best = s
    return ('ok', (out, (n, tot), best))
```

Functional

```
@doE
def _row(row):
    name, _, raw = row.partition(',')
    yield (Ok(name) if name else Err(f'empty name in {row!r}'))
    s = yield note(f'bad score in {row!r}', int(raw) if raw.strip().isdigit() else NOTHING)
    yield (Ok(s) if s <= 100 else Err(f'score > 100 in {row!r}'))
    return Ok((name, s))

stats = partial(foldMap, lambda p: ((1, p[1]), p[1]), both(both(Sum, Sum), MaxM))
admit_fn = lambda rows: bindE(traverseE(_row, rows),
                              lambda ps: Ok((ps, *stats(ps))))
```


Chapter 13

layers

layered config: defaults < file < cli; None is a real value

Imperative

```
_UNSET = object()                                # invent a private sentinel,
def effective_imp(layers, k):                      # because None must survive
    val = _UNSET
    for d in layers:
        if k in d:
            val = d[k]
    return NOTHING if val is _UNSET else val

def merged_imp(layers):
    out = {}
    for d in layers:
        for k, v in d.items():
            out[k] = v
    return out
```

Functional

```
effective_fn = lambda layers, k: mconcat>Last, [maybe_get(d, k) for d in layers])
merged_fn    = lambda layers: foldl(lambda a, b: unionWith(lambda old, new: new, a, b),
                                   layers, {})
```


Chapter 14

leaderboard

clamp scores into [0, 100], rank by score desc, name asc

Imperative

```
def rank_imp(rows):
    clamped = []
    for name, score in rows:
        if score < 0:
            score = 0
        elif score > 100:
            score = 100
        clamped.append((name, score))
    clamped.sort(key=lambda r: (-r[1], r[0]))      # score desc, name asc
    return clamped
```

Functional

```
clamp = lambda lo, hi: o(partial(max, lo), partial(min, hi))
clamp100 = clamp(0, 100)
name_of, score_of = itemgetter(0), itemgetter(1)
rank_fn = lambda rows: sortOn(
    lambda r: (-score_of(r), name_of(r)),
    [(name_of(r), clamp100(score_of(r))) for r in rows])
```


Chapter 15

logtrriage

'LEVEL message' lines: error count, total, first error message

Imperative

```
def triage_imp(lines):
    errors = 0                                # three mutable slots
    total = 0
    first = None                              # and a hand-rolled sentinel:
    seen_first = False                       # None alone can't mean "none yet"
    for ln in lines:                          # if a message could be empty
        lvl, _, msg = ln.partition(' ')
        total += 1
        if lvl == 'ERROR':
            errors += 1
            if not seen_first:
                first, seen_first = msg, True
    return ((errors, total), first if seen_first else NOTHING)
```

Functional

```
tag = lambda ln: (lambda lvl, _, msg:
    ((lvl == 'ERROR', 1), msg if lvl == 'ERROR' else NOTHING)
)(*ln.partition(' '))
triage_fn = partial(foldMap, tag, both(both(Sum, Sum), First))
```


Chapter 16

orderbatch

capstone: validate a batch of orders ($\text{qty} > 0$, $\text{price} > 0$), computing each line total; the whole batch fails at the first bad order, carrying which one and why

Imperative

```
def total_imp(order):
    if order["qty"] <= 0:
        return ("err", f"{order['item']}: bad qty {order['qty']}")
    if order["price"] <= 0:
        return ("err", f"{order['item']}: bad price {order['price']}")
    return ("ok", round(order["qty"] * order["price"], 2))

def admit_imp(orders):
    out = []
    for o in orders:
        r = total_imp(o)
        if r[0] == "err":
            return r
        out.append(r[1])
    return ("ok", out)
```

Functional

```
@doE
def total_fn(order):
    qty = yield (Ok(order["qty"]) if order["qty"] > 0
                 else Err(f"{order['item']}: bad qty {order['qty']}"))
    price = yield (Ok(order["price"]) if order["price"] > 0
                   else Err(f"{order['item']}: bad price {order['price']}"))
    return Ok(round(qty * price, 2))

admit_fn = lambda orders: sequenceE([total_fn(o) for o in orders])
```


Chapter 17

orgchart

resolve an employee's manager's manager's email, or NOTHING if any hop is missing. None-as-Nothing means a missing hop and a real missing-email must stay distinguishable

Imperative

```
def skip_manager_email_imp(emp):
    mgr = MANAGER_OF.get(emp, NOTHING)
    if mgr is NOTHING:
        return NOTHING
    skip = MANAGER_OF.get(mgr, NOTHING)
    if skip is NOTHING:
        return NOTHING
    if skip not in EMAIL_OF:
        return NOTHING
    return EMAIL_OF[skip]
```

Functional

```
@doM
def skip_manager_email_fn(emp):
    mgr = yield maybe_get(MANAGER_OF, emp)
    skip = yield maybe_get(MANAGER_OF, mgr)
    return (yield maybe_get(EMAIL_OF, skip))
```


Chapter 18

payoff

loan payoff schedule: balances after each fixed payment, and how many payments until paid off

Imperative

```
def schedule_imp(balance, payment, rate):
    out, n = [balance], 0
    while balance > 0:
        balance = round(balance * (1 + rate) - payment, 2)
        if balance < 0:
            balance = 0.0
        out.append(balance)
        n += 1
    return out, n
```

Functional

```
def _step(payment, rate):
    def step(balance):
        if balance <= 0:
            return NOTHING
        nxt = round(balance * (1 + rate) - payment, 2)
        nxt = max(nxt, 0.0)
        return (nxt, nxt)
    return step

def schedule_fn(balance, payment, rate):
    rest = unfoldr(_step(payment, rate), balance)
    return [balance] + rest, len(rest)
```


Chapter 19

sensoralert

smooth a sensor stream with a 3-wide moving average, then report each consecutive run of over-threshold readings as (start, len)

Imperative

```
def alerts_imp(xs, n, threshold):
    smoothed = []
    for i in range(len(xs) - n + 1):
        window = xs[i:i + n]
        smoothed.append(sum(window) / n)
    flags = [v > threshold for v in smoothed]
    runs, i = [], 0
    while i < len(flags):
        if flags[i]:
            start = i
            while i < len(flags) and flags[i]:
                i += 1
            runs.append((start, i - start))
        else:
            i += 1
    return runs
```

Functional

```
def alerts_fn(xs, n, threshold):
    smoothed = [sum(w) / n for w in windows(n, xs)]
    flagged = [(i, v > threshold) for i, v in enum(smoothed)]
    groups = groupBy(lambda a, b: a[1] == b[1], flagged)
    return [(g[0][0], len(g)) for g in groups if g[0][1]]
```


Chapter 20

wordfreq

top-N word frequencies, ties alphabetical

Imperative

```
def top_imp(n, text):
    counts = {}
    for w in text.lower().split():
        if w in counts:
            counts[w] += 1
        else:
            counts[w] = 1
    items = list(counts.items())
    items.sort(key=lambda kv: (-kv[1], kv[0]))
    return items[:n]
```

mutate a dict

then sort, then slice

Functional

```
top_fn = lambda n, text: pipe(
    text, str.lower, words,
    lambda ws: fromListWith(add, [(w, 1) for w in ws]),
    dict.items, partial(sortOn, lambda kv: (-kv[1], kv[0])),
    partial(take, n))
```


Part III

Appendices

Appendix A

haskell.py, Complete

```
1  """haskell.py -- the comprehensive Haskell-flavoured toolbox for Python.
2
3  One importable vocabulary: the working parts of the Prelude, Data.List,
4  Data.Maybe, Data.Either, Data.Map, Data.Function, Control.Monad, monoids,
5  and parser combinators, folded into sections and mastered as one language.
6
7  Design rules
8      * Where the stdlib already has it, the Haskell name WRAPS the stdlib
9        (scanl is itertools.accumulate, memo is functools.cache) -- one
10       vocabulary, zero reimplementation.
11      * Failure is a value. Maybe uses the true NOTHING sentinel, so None is
12        LEGAL data everywhere; parsers fail with FAIL, and Either carries why.
13      * do-notation via generators: `yield` is >=> (linear monads only).
14      * Constant stack wherever Haskell would lean on TCO: loops, not recursion.
15      * Python keywords and builtins keep prelude.py's underscore convention
16        (not_, map_, filter_, zip_); `id` stays Python's -- use identity.
17  """
18  from functools import reduce as _reduce, partial, cache as memo, cmp_to_key
19  from itertools import accumulate as _acc, islice as _islice, pairwise as _pairwise
20  from itertools import chain, count, cycle, repeat
21  from operator import add, sub, mul, not_
22  from math import gcd, lcm, hypot, isqrt, prod as product
23  from collections import Counter, defaultdict, deque
24  from heapq import heappush, heappop, merge as _hmerge
25  import re as _re
26
27  # ===== 1. functions =====
28  identity = lambda x: x
29  const    = lambda x: lambda _: x
30  flip     = lambda f: lambda a, b: f(b, a)
31  curry    = lambda f: lambda x: lambda y: f(x, y)
32  uncurry  = lambda f: lambda p: f(*p)
33
34  def compose(*fs):
35      """Right-to-left composition, iterative (constant stack).
36      >>> compose(str, abs)(-3)
37      '3'
38      >>> o(str, abs)(-3)
39      '3'
40      """
41      def g(x):
42          for f in reversed(fs):
43              x = f(x)
44          return x
45      return g
46
47  o = compose                # ML's operator: o(f, g) is f after g
48
49  def pipe(x, *fs):
50      """Left-to-right application: pipe(x, f, g) == g(f(x)).
51      >>> pipe(-3, abs, str)
52      '3'
53      """
54      for f in fs:
```

```

55     x = f(x)
56     return x
57
58 def on(f, g):
59     """Data.Function: combine two arguments via a key.
60     >>> on(sub, len)('haskell', 'py')
61     5
62     """
63     return lambda x, y: f(g(x), g(y))
64
65 # ===== 2. pairs =====
66 fst = lambda p: p[0]
67 snd = lambda p: p[1]
68 swap = o(tuple, reversed)
69
70 # ===== 3. sentinels =====
71 class _Sentinel:
72     __slots__ = ("_n",)
73     def __init__(self, n): self._n = n
74     def __repr__(self): return self._n
75     def __bool__(self): return False
76
77 NOTHING = _Sentinel("NOTHING") # Maybe's Nothing; None is a legal Just here
78 FAIL = _Sentinel("FAIL") # parser failure; (None, rest) is a legal success
79 _MISS = _Sentinel("_MISS") # internal: absent optional argument
80
81 # ===== 4. arithmetic & logic =====
82 # add, sub, mul, not_ are operator's; gcd, lcm, hypot, isqrt, product are math's
83 div = lambda a, b: a // b # floors, like Haskell div
84 mod = lambda a, b: a % b # sign follows b
85 halve = lambda n: n // 2
86 even = lambda n: n % 2 == 0
87 odd = lambda n: n % 2 == 1
88 otherwise = True
89
90 def quot(a, b):
91     """Truncating division, Haskell quot; rem's sign follows a.
92     >>> quotRem(-7, 2)
93     (-3, -1)
94     """
95     return -(-a // b) if (a < 0) != (b < 0) else a // b
96 rem = lambda a, b: a - b * quot(a, b)
97 quotRem = lambda a, b: (quot(a, b), rem(a, b))
98 signum = lambda x: (x > 0) - (x < 0)
99 succ = lambda x: chr(ord(x) + 1) if isinstance(x, str) else x + 1
100 pred = lambda x: chr(ord(x) - 1) if isinstance(x, str) else x - 1
101
102 # ===== 5. folds, scans, unfolds =====
103 def foldl(f, xs, base=_MISS):
104     """foldl; seeds with the first element if no base is given.
105     >>> foldl(lambda a, b: a - b, [1, 2, 3], 10)
106     4
107     """
108     return _reduce(f, xs) if base is _MISS else _reduce(f, xs, base)
109
110 def foldr(f, xs, base):
111     """Right fold without recursion.
112     >>> foldr(lambda x, acc: [x] + acc, [1, 2, 3], [])
113     [1, 2, 3]
114     """
115     return _reduce(flip(f), reversed(list(xs)), base)
116
117 foldl1 = foldl
118 foldr1 = lambda f, xs: _reduce(flip(f), reversed(list(xs)))
119
120 def scanl(f, xs, base=_MISS):
121     """All intermediate foldl states (itertools.accumulate).
122     >>> scanl(add, [1, 2, 3], 0)
123     [0, 1, 3, 6]
124     """
125     return list(_acc(xs, f) if base is _MISS else _acc(xs, f, initial=base))
126
127 scanl1 = lambda f, xs: scanl(f, xs)
128 scanr = lambda f, xs, base: scanl(flip(f), reversed(list(xs)), base)[::-1]

```

```

129 scanr1 = lambda f, xs: scanl(flip(f), reversed(list(xs)))[:-1]
130
131 def unfoldr(f, seed):
132     """Anamorphism: f(seed) -> (value, seed') or NOTHING to stop.
133     >>> unfoldr(lambda n: NOTHING if n == 0 else (n, n - 1), 4)
134     [4, 3, 2, 1]
135     """
136     out = []
137     while (r := f(seed)) is not NOTHING:
138         out.append(r[0]); seed = r[1]
139     return out
140
141 def until(cond, f, x):
142     """Loop f until cond holds.
143     >>> until(lambda n: n > 100, lambda n: n * 2, 1)
144     128
145     """
146     while not cond(x):
147         x = f(x)
148     return x
149
150 # brute-force enumerations (say the bound out loud):
151 replicateM = lambda n, xs: foldl(lambda acc, _: [s + [x] for s in acc for x in xs], range(n), [[]]) # |xs|^n
152 subsequences = lambda xs: foldl(lambda acc, x: acc + [s + [x] for s in acc], list(xs), [[]]) # 2^n
153
154 # ===== 6. streams ===== (lazy; chain, count, cycle, repeat are itertools')
155 def iterate(f, x):
156     """Infinite stream x, f(x), f(f(x)), ...
157     >>> take(5, iterate(lambda n: n * 2, 1))
158     [1, 2, 4, 8, 16]
159     """
160     while True:
161         yield x
162         x = f(x)
163
164 take = lambda n, xs: list(_islice(iter(xs), n)) # safe on infinite streams
165
166 # ===== 7. list basics =====
167 head = lambda xs: xs[0]
168 tail = lambda xs: xs[1:]
169 init = lambda xs: xs[:-1]
170 last = lambda xs: xs[-1]
171 drop = lambda n, xs: list(xs)[max(n, 0):]
172 splitAt = lambda n, xs: (xs[:n], xs[n:]) if n >= 0 else (xs[:0], xs)
173 replicate = lambda n, x: [x] * n
174 elem = lambda x, xs: x in xs
175 notElem = lambda x, xs: x not in xs
176 null = lambda xs: len(xs) == 0
177 nub = o(list, dict.fromkeys) # unique, first seen wins
178 enum = lambda xs, start=0: list(enumerate(xs, start))
179 pairwise = lambda xs: list(_pairwise(xs))
180 zip_ = lambda a, b: list(zip(a, b))
181 zip3 = lambda a, b, c: list(zip(a, b, c))
182 unzip = lambda ps: tuple(map(list, zip(*ps))) if ps else ([], [])
183 isPrefixOf = lambda p, xs: list(xs[:len(p)]) == list(p)
184 isSuffixOf = lambda p, xs: list(xs[len(xs) - len(p):]) == list(p)
185
186 def lookup(k, pairs):
187     """Assoc-list lookup into Maybe (a stored None is data).
188     >>> lookup('b', [('a', 1), ('b', None)]) is None
189     True
190     >>> lookup('c', [('a', 1)]) is NOTHING
191     True
192     """
193     return next((v for kk, v in pairs if kk == k), NOTHING)
194
195 def stripPrefix(p, xs):
196     """The rest after prefix p, or NOTHING.
197     >>> stripPrefix('foo', 'foobar')
198     'bar'
199     >>> stripPrefix('x', 'foobar') is NOTHING
200     True
201     """
202     return xs[len(p):] if isPrefixOf(p, xs) else NOTHING

```

```

203
204 # ===== 8. higher-order lists =====
205 map_      = lambda f, xs: [f(x) for x in xs]
206 filter_   = lambda cond, xs: [x for x in xs if cond(x)]
207 concat    = lambda xss: [x for xs in xss for x in xs]
208 concatMap = lambda f, xs: [y for x in xs for y in f(x)]
209 cross     = lambda a, b: [(x, y) for x in a for y in b]
210 starmap   = lambda f, pairs: [f(*p) for p in pairs]
211 zipWith   = lambda f, a, b: [f(x, y) for x, y in zip(a, b)]
212 zipWith3  = lambda f, a, b, c: [f(x, y, z) for x, y, z in zip(a, b, c)]
213
214 def find(cond, xs):
215     """First match, lazily, into Maybe.
216     >>> find(even, [1, 3, 4, 5])
217     4
218     >>> find(even, [1, 3]) is NOTHING
219     True
220     """
221     return next((x for x in xs if cond(x)), NOTHING)
222
223 def partition(cond, xs):
224     """(keepers, rest) in ONE pass, cond called once per element.
225     >>> partition(even, [1, 2, 3, 4])
226     ([2, 4], [1, 3])
227     """
228     yes, no = [], []
229     for x in xs:
230         (yes if cond(x) else no).append(x)
231     return (yes, no)
232
233 # ===== 9. slicing & spans =====
234 def span(cond, xs):
235     """Split at first failure, one pass, one cond call per element.
236     >>> span(lambda x: x < 3, [1, 2, 3, 1])
237     ([1, 2], [3, 1])
238     """
239     xs = list(xs)
240     i = next((i for i, x in enumerate(xs) if not cond(x)), len(xs))
241     return (xs[:i], xs[i:])
242
243 break_ = lambda cond, xs: span(o(not_, cond), xs)
244
245 def takewhile(cond, xs):                # generator
246     for x in xs:
247         if not cond(x):
248             return
249         yield x
250
251 def dropwhile(cond, xs):                # generator
252     it = iter(xs)
253     for x in it:
254         if not cond(x):
255             yield x
256             break
257     yield from it
258
259 def takeuntil(cond, xs):                # takewhile that INCLUDES the first hit
260     for x in xs:
261         yield x
262         if cond(x):
263             return
264
265 takeWhile = lambda cond, xs: list(takewhile(cond, xs))
266 dropWhile = lambda cond, xs: list(dropwhile(cond, xs))
267 inits     = lambda xs: [xs[:i] for i in range(len(xs) + 1)]
268 tails    = lambda xs: [xs[i:] for i in range(len(xs) + 1)]
269
270 def chunksOf(n, xs):
271     """
272     >>> chunksOf(2, [1, 2, 3, 4, 5])
273     [[1, 2], [3, 4], [5]]
274     """
275     xs = list(xs)
276     return [xs[i:i + n] for i in range(0, len(xs), n)]

```

```

277
278 def windows(n, xs):
279     """Sliding windows of width n.
280     >>> windows(3, [1, 2, 3, 4])
281     [[1, 2, 3], [2, 3, 4]]
282     """
283     xs = list(xs)
284     return [xs[i:i + n] for i in range(len(xs) - n + 1)]
285
286 def groupBy(eq, xs):
287     """Haskell semantics: runs anchored on each run's FIRST element.
288     >>> groupBy(lambda a, b: a == b, "aabba")
289     [['a', 'a'], ['b', 'b'], ['a']]
290     """
291     out = []
292     for x in xs:
293         if out and eq(out[-1][0], x):
294             out[-1].append(x)
295         else:
296             out.append([x])
297     return out
298
299 group = partial(groupBy, lambda a, b: a == b)
300
301 def transpose(rows):
302     """Ragged-safe like Data.List.
303     >>> transpose([[1, 2, 3], [4, 5]])
304     [[1, 4], [2, 5], [3]]
305     """
306     rows = [list(r) for r in rows]
307     out, i = [], 0
308     while any(i < len(r) for r in rows):
309         out.append([r[i] for r in rows if i < len(r)])
310         i += 1
311     return out
312
313 # ===== 10. sorting & searching =====
314 sortOn = lambda f, xs: sorted(xs, key=f)
315 sortBy = lambda cmp, xs: sorted(xs, key=cmp_to_key(cmp))
316 maxOn = lambda f, xs: max(xs, key=f)
317 minOn = lambda f, xs: min(xs, key=f)
318 merge = lambda a, b: list(_hmerge(a, b))          # stable merge of sorted lists
319
320 # ===== 11. strings =====
321 lines = str.splitlines
322 unlines = lambda ls: "".join(l + "\n" for l in ls)
323 words = str.split
324 unwords = " ".join
325
326 # ===== 12. containers ===== (Counter, defaultdict, deque, heappush/heappop are stdlib's)
327 def fromListWith(f, pairs):
328     """Data.Map: THE dict-building fold; f(new, old).
329     >>> fromListWith(add, [('a', 1), ('b', 2), ('a', 3)]) == {'a': 4, 'b': 2}
330     True
331     """
332     d = {}
333     for k, v in pairs:
334         d[k] = f(v, d[k]) if k in d else v
335     return d
336
337 def insertWith(f, k, v, d):
338     """Persistent insert (fresh dict); f(new, old).
339     >>> insertWith(add, 'a', 5, {'a': 1}) == {'a': 6}
340     True
341     """
342     return {**d, k: f(v, d[k]) if k in d else v}
343
344 def unionWith(f, a, b):
345     """Data.Map: merge two dicts, combining shared keys with f.
346     >>> unionWith(add, {'a': 1, 'b': 2}, {'b': 3, 'c': 4}) == {'a': 1, 'b': 5, 'c': 4}
347     True
348     """
349     return {k: f(a[k], b[k]) if k in a and k in b else (a[k] if k in a else b[k])
350             for k in a.keys() | b.keys()}

```

```

351
352 def getpath(t, ks, default=NOTHING):
353     """Read along a key path without autovivifying; Maybe out.
354     >>> getpath({'a': {'b': None}}, ['a', 'b']) is None
355     True
356     >>> getpath({'a': {}}, ['a', 'b']) is NOTHING
357     True
358     """
359     for k in ks:
360         if not isinstance(t, dict) or k not in t:
361             return default
362         t = t[k]
363     return t
364
365 # multiset ops on count-dicts (use .get -- indexing a Counter autovivifies):
366 bag_diff = lambda a, b: {k: a[k] - b.get(k, 0) for k in a if a[k] > b.get(k, 0)}
367 bag_inter = lambda a, b: {k: v for k in a.keys() & b.keys() if (v := min(a[k], b[k]))}
368 bag_union = lambda a, b: {k: max(a.get(k, 0), b.get(k, 0)) for k in a.keys() | b.keys()}
369 bag_sub = lambda a, b: all(b.get(k, 0) >= n for k, n in a.items())
370
371 # ===== 13. Maybe ===== (NOTHING sentinel; None is a legal value)
372 def bind(x, f):
373     """>>= : short-circuit on NOTHING.
374     >>> bind(3, lambda v: v + 1), bind(NOTHING, lambda v: v + 1)
375     (4, NOTHING)
376     """
377     return NOTHING if x is NOTHING else f(x)
378
379 isJust = lambda x: x is not NOTHING
380 isNothing = lambda x: x is NOTHING
381 fromMaybe = lambda default, x: default if x is NOTHING else x
382
383 listToMaybe = lambda xs: xs[0] if xs else NOTHING
384 maybeToList = lambda x: [] if x is NOTHING else [x]
385
386 def catMaybes(xs):
387     """
388     >>> catMaybes([1, NOTHING, None, 2])
389     [1, None, 2]
390     """
391     return [x for x in xs if x is not NOTHING]
392
393 def mapMaybe(f, xs):
394     """Map into Maybe and drop the NOTHINGS, one pass.
395     >>> mapMaybe(lambda s: int(s) if s.isdigit() else NOTHING, ['3', 'x', '7'])
396     [3, 7]
397     """
398     return [y for y in map(f, xs) if y is not NOTHING]
399
400 def sequenceM(xs):
401     """All-or-nothing.
402     >>> sequenceM([1, 2]), sequenceM([1, NOTHING])
403     ([1, 2], NOTHING)
404     """
405     out = []
406     for x in xs:
407         if x is NOTHING:
408             return NOTHING
409         out.append(x)
410     return out
411
412 traverseM = lambda f, xs: sequenceM(map_(f, xs))
413
414 def maybe_get(d, k):
415     """dict lookup into Maybe (None stays distinguishable from missing).
416     >>> maybe_get({'a': None}, 'a'), maybe_get({}, 'a')
417     (None, NOTHING)
418     """
419     return d.get(k, NOTHING)
420
421 # ===== 14. Either ===== (tagged tuples; match-friendly)
422 Ok = lambda v: ("ok", v)
423 Err = lambda e: ("err", e)
424

```



```

425 def bindE(x, f):
426     """
427     >>> bindE(Ok(3), lambda v: Ok(v + 1)), bindE(Err("no"), lambda v: Ok(v))
428     (('ok', 4), ('err', 'no'))
429     """
430     return x if x[0] == "err" else f(x[1])
431
432 def sequenceE(xs):
433     """First Err wins; else Ok of all values.
434     >>> sequenceE([Ok(1), Err("a"), Err("b")])
435     ('err', 'a')
436     """
437     out = []
438     for x in xs:
439         if x[0] == "err":
440             return x
441         out.append(x[1])
442     return Ok(out)
443
444 def traverseE(f, xs):
445     """Map into Either, then sequence.
446     >>> traverseE(lambda n: Ok(n * 2) if n else Err('zero'), [1, 2])
447     ('ok', [2, 4])
448     """
449     return sequenceE(map_(f, xs))
450
451 def partitionEithers(xs):
452     """
453     >>> partitionEithers([Ok(1), Err('a'), Ok(2)])
454     ([1, 2], ['a'])
455     """
456     oks, errs = [], []
457     for tag, v in xs:
458         (oks if tag == "ok" else errs).append(v)
459     return (oks, errs)
460
461 def note(msg, x):
462     """Maybe -> Either: attach WHY to a NOTHING.
463     >>> note("missing key", NOTHING)
464     ('err', 'missing key')
465     """
466     return Err(msg) if x is NOTHING else Ok(x)
467
468 # ===== 15. do-notation ===== (yield is >>=; linear monads only)
469 def do(binder, is_fail):
470     """Build a do-block decorator for any short-circuiting monad."""
471     def deco(gen_fn):
472         def run(*a, **kw):
473             g = gen_fn(*a, **kw)
474             def step(val):
475                 try:
476                     m = g.send(val)
477                     except StopIteration as e:
478                         return e.value
479                     return binder(m, step)
480             return step(None)
481         return run
482     return deco
483
484 doM = do(bind, lambda x: x is NOTHING) # generator returns a plain value
485 doE = do(bindE, lambda x: x[0] == "err") # generator returns Ok(...)/Err(...)
486
487 # ===== 16. parser combinators =====
488 # Parser a = str -> (value, rest) | FAIL. Values may be None (json null!).
489 def doP(gen_fn):
490     """do-notation that THREADS the input: yield a parser, receive its value.
491     Fresh generator per parse -- re-runnable, backtrack-safe under alt."""
492     def make(*a, **kw):
493         def parser(s):
494             g, val = gen_fn(*a, **kw), None
495             try:
496                 while True:
497                     r = g.send(val)(s)
498                     if r is FAIL:

```

```

499         return FAIL
500         val, s = r
501     except StopIteration as e:
502         return (e.value, s)
503     return parser
504 return make
505
506 def alt(*ps):
507     """</> : first success wins."""
508     return lambda s: next((r for r in (p(s) for p in ps) if r is not FAIL), FAIL)
509
510 def many(p):
511     """Zero or more, loop not recursion; stalls guard against empty matches."""
512     def parser(s):
513         out = []
514         while (r := p(s)) is not FAIL and r[1] != s:
515             out.append(r[0]); s = r[1]
516         return (out, s)
517     return parser
518
519 def many1(p):
520     def parser(s):
521         r = p(s)
522         if r is FAIL:
523             return FAIL
524         rest = many(p)(r[1])
525         return ([r[0]] + rest[0], rest[1])
526     return parser
527
528 def sepBy(p, sep):
529     def pair(s):
530         r = sep(s)
531         return FAIL if r is FAIL else p(r[1])
532     def parser(s):
533         r = p(s)
534         if r is FAIL:
535             return ([], s)
536         rest = many(pair)(r[1])
537         return ([r[0]] + rest[0], rest[1])
538     return parser
539
540 def lit(c):
541     """One expected character, whitespace-blind."""
542     def parser(s):
543         s = s.lstrip()
544         return (c, s[1:]) if s[:1] == c else FAIL
545     return parser
546
547 def rx(pat, conv=identity):
548     """Regex token, whitespace-blind, converted."""
549     r = _re.compile(pat)
550     def parser(s):
551         s = s.lstrip()
552         m = r.match(s)
553         return (conv(m.group()), s[m.end():]) if m else FAIL
554     return parser
555
556 def chainl1(p, ops):
557     """p (op p)* folded LEFT; ops maps operator char -> binary function."""
558     opp = alt(*[lit(c) for c in ops])
559     def parser(s):
560         vr = p(s)
561         while vr is not FAIL:
562             op = opp(vr[1])
563             if op is FAIL:
564                 return vr
565             w = p(op[1])
566             if w is FAIL:
567                 return vr
568             vr = (ops[op[0]](vr[0], w[0]), w[1])
569         return FAIL
570     return parser
571
572 def runParser(p, s):

```

```

573     """Parser -> Either: Ok(value) or Err showing where it stopped.
574     >>> runParser(rx(r'\d+', int), "42")
575     ('ok', 42)
576     >>> runParser(rx(r'\d+', int), "42x")
577     ('err', "unconsumed input: 'x'")
578     """
579     r = p(s)
580     if r is FAIL:
581         return Err(f"no parse: {s!r}")
582     if r[1].strip():
583         return Err(f"unconsumed input: {r[1].strip()!r}")
584     return Ok(r[0])
585
586 # ===== 17. monoids =====
587 Monoid = lambda empty, op: (empty, op)
588 mconcat = lambda m, xs: foldl(m[1], xs, m[0])
589 foldMap = lambda f, m, xs: mconcat(m, [f(x) for x in xs])
590 both = lambda m1, m2: Monoid((m1[0], m2[0]),
591                               lambda a, b: (m1[1](a[0], b[0]), m2[1](a[1], b[1])))
592 Sum = Monoid(0, add)
593 Product = Monoid(1, mul)
594 All = Monoid(True, lambda a, b: a and b)
595 Any = Monoid(False, lambda a, b: a or b)
596 MaxM = Monoid(float("-inf"), max)
597 MinM = Monoid(float("inf"), min)
598 ListM = Monoid([], add)
599 First = Monoid(NOTHING, lambda a, b: b if a is NOTHING else a)
600 Last = Monoid(NOTHING, lambda a, b: a if b is NOTHING else b)
601
602 if __name__ == "__main__":
603     import doctest
604     fails, total = doctest.testmod()
605     print(f"{total - fails}/{total} doctests passed" if not fails
606           else f"{fails} FAILED")

```

